TrES-2b

R-0158

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-2_180906` · created 2026-07-16T15:05:08+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458367.67934 +/- 0.0006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1489 +/- 0.0091
Transit depth (Rp/Rs)^2	2.22 +/- 0.27 [%]
Orbital Inclination (inc)	83.75 +/- 0.33
Airmass coefficient 1 (a1)	1.045 +/- 0.012
Airmass coefficient 2 (a2)	-0.038 +/- 0.031
Scatter in the residuals of the lightcurve fit is	1.1 %
Best Comparison Star	#1 - [484, 123]
Optimal Aperture	4.7
Optimal Annulus	11.63
Transit Duration (day)	0.0761 +/- 0.005

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1489 ± 0.0091 vs 0.1254 (deviation 2.36σ)
Duration fit vs published	0.0761 d vs 0.0746 d (deviation 0.28σ)
Mid-time O–C	-1.5 min
Depth SNR	14.9
Residual scatter	1.1 %

Light curve

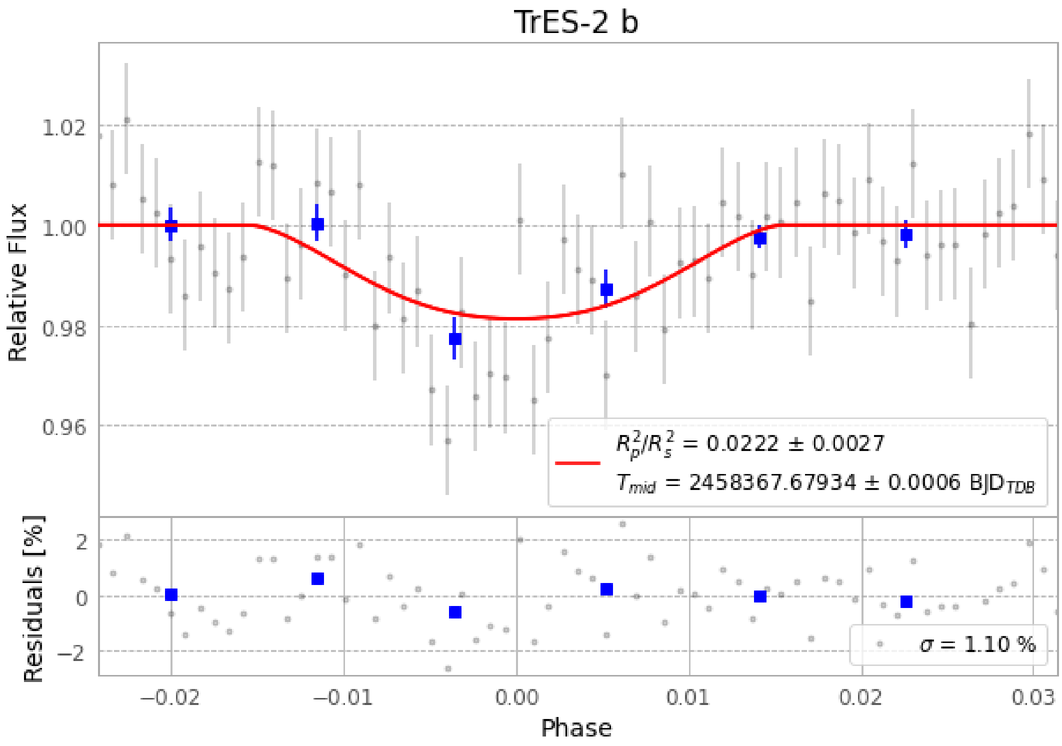

Figure 1 — FinalLightCurve_TrES-2 b_2018-09-05.png (SHA-256 4633a7a5f2b56688…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [389, 287], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 11e2632728c8ff6a…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit c71eff8`

Data provenance

70 FITS frames (46 MB), `TRES-2180906023912.FITS` → `TRES-2180906060612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-2-180906.log (dfac4b82802a…), FinalLightCurve_TrES-2 b_2018-09-05.png (4633a7a5f2b5…), FinalLightCurve_TrES-2 b_2018-09-05.csv (ddf9210f3045…)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)